Physics 11 Practice: Forces I

Name:____________________________

Date:____________________ **Block:**____

force = mass x gravitational field strength; F = ma; $F_g = mg$; $F_F = \mu F_N$; $F_e = kx$
g = earth gravitational field strength vs. acceleration due to gravity = 9.8 N/kg vs m/s^2*[down]*

1. Find force of gravity acting on a: a) 1500 kg truck b) 4.5 kg block of concrete

2. a) Find mass of an object being acted on by a force of gravity of 167 N at earth's surface

   b) Find the gravitational field strength of the moon if the force of gravity on a 250 kg spacecraft on the moon is 408 N

3. a) 50 N used to pull a 6.0 kg object along a table at constant speed. Find coefficient of friction.

   b) Coefficient of friction between two materials is 0.35. A 5.0 kg object made of one of the materials is pulled along a surface made of other material. Find force of friction.

   c) A tractor is pulling a plough through a field with a force of 880 N. Plough has a mass of 100 kg and is moving at a constant speed. Find coefficient of friction.

4. a) A metallic spring has a spring constant of 40 N/m. How much force would it take to stretch the spring by 10 cm?

   b) A garage door has a spring constant of 600 N/m. How much will door stretch if a force of 150 N is applied to it?

   c) Find spring constant of a car's shock absorber, if a 2500 N force compresses shock absorber from a length of 50 cm to a length of 40 cm.

   d) A block of wood with a spring constant of 1.4×10^6 N/m broke after a force of 6.0×10^3 N was applied. If the starting position of the block was 1.5 mm, what position did the block stretch to just before it broke?

5. a) Find net force needed to accelerate a 5.0 kg object at 2.0 m/s^2

   b) Find acceleration produced when a net force of 12.0 N acts on 0.030 kg ball

   c) Find mass of a a ball that accelerates at 4.0×10^3 m/s^2 when hit with a force of 1000 N.

   d) Find acceleration produced when a net force of 20 N acts on 5.0 kg ball

   e) Find mass of an object that that accelerates at 10 m/s^2 due to a net force of 150 N

6. a) A car with a mass of 1200 kg is pushed along a level road (force of friction on road = 500 N) with a thrust of 700 N. Find acceleration of car.

   b) 3.0 kg mass is pulled along a level table top by a force of 16 N and it accelerates at 4.0 m/s^2. Find coefficient of friction between object/table top.

   c) A box is pulled along a level surface (coefficient of friction for box/surface = 0.52) by a force of 650 N, causing box to accelerate at 2.8 m/s^2. Find mass of box.

7. a) Rocket with mass = 2.92 x 10^6 kg has engines that produce a thrust of 2.4 x 10^8 N.
   Find: i) downward force (F_g) of rocket
   ii) net force acting on rocket
   iii) acceleration of rocket
   iv) Will acceleration of rocket stay constant as engines continue to fire? Explain
   (What happens to rocket as fuel is burned?)

   b) i) Calculate initial acceleration of a 13,140 kg rocket if thrust of its engines is 2.63 x 10^5 N.

   ii) Then calculate rocket's acceleration near "burn-out" when its mass is only 4170 kg.

8. a) A cannonball with a mass of 5.0 kg is shot horizontally in a tube (no projectile motion), accelerating from rest to 50 m/s in 0.050 s. Find net force acting on cannonball.

   b) How long would it take a 50 kg rider on a 10 kg bike to accelerate from rest to 4.0 m/s on level ground if net force acting on the bike is 48 N?

   c) A 5.0 kg stone is sinking in water at a constant speed.
   Find upward (buoyancy) force of water on stone

   d) A net force of 2.5×10^3 N acts on an object for 5.0 s.
   During this time the object accelerates from rest to a velocity of 48 km/h.
   Find the mass of this object.

   e) A net force of 6.6 N acts on a 9.0 kg object, causing it to accelerate from rest to a velocity of 3.0 m/s: i) How far did the object travel while accelerating?
   ii) Find the time of acceleration

KEY:

1. a) 15000 N	b) 44 N			
2. a) 17.0 kg	b) 1.6 N/kg			
3. a) 0.85	b) 17 N	c) 0.9		
4. a) 4 N	b) 0.3 m	c) 25000 N/m	d) 5.8 mm	
5. a) 1.0 x 10^{1} N	b) 4.0 x 10^2 m/s^2	c) 0.3 kg	d) 4 m/s^2	e) 15 kg
6. a) 0.2 m/s^2	b) 0.14	c) 82 kg		
7. a) i) 2.9 x 10^7 N	ii) 2.1 x 10^8 N	iii) 72 m/s^2	iv) no, using fuel; ↓ m ↑ a	
b) i) 10.2 m/s^2	ii) 53.3 m/s^2			
8. a) 5000 N	b) 5 s	c) 49 N	d) 940 kg	e) i) 6.1 m ii) 4.1 s

Explanation Questions:

9. Use Newton's law of motion to….

   a) Explain the situations below:

      i) A jet is travelling at a constant speed in a straight line, describe the interaction of forces acting on jet
      ii) A person is pushing a chair across the floor at a constant speed, Describe how applied force is related to force of friction
      iii) Truck accelerates @ 0.3 m/s^2 - describe interaction of forces acting on truck
      iv) Passenger does not have seat belt buckled when car crashes into pole
      v) Car attempts to stop at a traffic light on an icy street
      vi) Passenger sips a coffee just as turbulence causes a plane to drop 3 m

   b) Answer the questions below:
      i) As you go <u>up</u> stairs, in which direction do your feet push?
      ii) As you swim <u>forward</u>, in which direction must your arms push?
      iii) What other forces are acting as a ball is kicked upward with an applied force?
      iv) Why does a tennis ball bounce back up when you throw it to the ground?
      v) Explain how a person stranded, motionless in the center of a frictionless ice rink can get to edge of rink and escape

   c) Complete the statements below:

      i) As you row a boat <u>forward</u>, your oars push…
      ii) As a rocket flies <u>up</u>, its, engines fire….
      iii) As you exert a force on the ground [downwards] while you walk…
      iv) As a jet exerts a force on the air while flying…
      v) As a canoe paddle applies an Eastward force on the water…
      vi) As wheels of a car push on road with a force of 250 N [South] & 2500 N [down]…

KEY

9. a) i) $F_{net} = 0$ (balanced F)
   ii) $F_A = F_F$ ($F_{net} = 0$)
   iii) unbalanced forces: since truck accelerating (thrust > friction; $F_A > F_F$)
   iv) pole applies force to stop car; person has inertia and is not attached to car, person moves forward at speed car was going
   v) coefficient of friction between car and icy road not enough for force of friction to stop car; car's inertia causes it to maintain initial speed/direction
   vi) plane & person are "attached", both drop 3 m, coffee's inertia means it will only drop with gravity, coffee appears to stay still or move upward slightly as plane drops

   b) i) down
   ii) backward
   iii) gravity, air resistance
   i) ground pushes up on ball
   ii) throw something heavy, blow air (action/reaction)

   c) i) backward
   ii) down
   iii) ground exerts a force on you (upwards)
   iv) air exerts force on jet
   v) water exerts force = applied forceon paddle
   vi) road exerts a force of 250 N [North] & 2500 N [up] on wheels